Can we identify properties that predict whether a fact will be retained versus forgotten?

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Motivation

Fact frequency is a well-established driver of factual learning in language models. LMEnt (Gottesman et al., 2025) reports that as a fact's frequency increases, both its learning and forgetting rates rise during pretraining, leaving the cause open. We ask: holding frequency fixed, which properties of the corpus and training schedule predict whether a fact is retained versus forgotten? Understanding which factual knowledge language models retain, and why some is later lost, matters both for understanding learning dynamics and for controlling knowledge acquisition during training. Conclusions from this research could inform data-ordering choices during pretraining.

Workflow

Step A - Data generation: We would use three sources of data, with their raw values and engineered features:

1. Acquisition timeline: We iterate over the model's checkpoints and test recall for each fact*. Derive additional per-fact features as first-learned step, number of forgetting events, and oscillation count.
2. Training data schedule: We map when each fact's associated chunks were presented to the model, and extract temporal features such as within-epoch spacing patterns and cumulative appearance counts up to each checkpoint.
3. Corpus content: fact's relation type, subject chunk counts, answer chunks count, shared chunk count.

Step B - Exploratory data analysis: holding frequency fixed, we would try to identify features that discriminate retained from forgotten facts. Manual analysis to build intuition, look at distributions, and spot confounds. Test trivial assumptions like the fact's subject and first learned step.

Step C - Prediction model: Train an interpretable prediction model (regression or tree-based) on the data (or subsets of the features each time). Then investigate each feature's effect *at fixed frequency* via coefficients, SHAP, or partial-dependence plots.

(Optional) Step D - Experimental validation: Based on the results, we can design a pretraining scheme to experimentally validate our findings. Even a negative result such as "spacing doesn't matter" is valuable and can be tested.

Assumptions and requirements

- **Models/data:** LMEnt-1B models (1-epoch**), with their intermediate checkpoints, the entity-annotated corpus, the Elasticsearch index, and the annotated PopQA data - all public on HuggingFace.
- **Compute:** Slurm studentkillable partition; one 40–80GB GPU for checkpoint-evaluation sweeps (inference only). Index queries run CPU-side.
- **Storage:** Tens of GB for cached eval outputs and per-interval feature tables.

References

Gottesman, Daniela, et al. "LMEnt: A suite for analyzing knowledge in language models from pretraining data to representations." *Sep 2025.* https://arxiv.org/abs/2509.03405

Merullo, Jack, et al. "On Linear Representations and Pretraining Data Frequency in Language Models." *ICLR 2025.* https://arxiv.org/abs/2504.12459

* Instead of recording only the binary substring-match label per fact per checkpoint, we can also extract the gold-answer probability, giving us a more fine-grained signal.

** We chose to start with the 1-epoch model because it is the one the paper focuses on, and for simplicity, but the multi-epoch models could also be interesting.

Plan B:

- Instead of drawing conclusions from the observational data, we could design different epoch schedules and evaluate their effect. For example, batches stratified by relation type versus batches where all chunks share the same relation type. (This requires pretraining runs, so it is heavier than the main plan.)

- Lightly fine-tune on the forgotten facts and study the resulting dynamics: can we characterize which facts get forgotten as new ones are learned (knowledge replacement)? Are they concentrated in a specific domain? Are they the most recently or least recently learned?

- Re-run the whole experiment several times with different random orderings of the epoch, and check whether the same facts are consistently forgotten. Consistency would suggest forgetting is a property of the fact, while variability would point to the schedule.